[01_IDENTITY]

S Madhan Kumar

A highly skilled Data Analyst and Lead Developer with a Bachelor's in Computer Science specializing in Data Science. Proficient in developing end-to-end analytical pipelines, optimizing workflows, and synthesizing critical business intelligence. Experienced in leveraging Python, Java, SQL, and various AI/ML frameworks to build scalable solutions, including a patent-pending framework for environmental analytics and AI-powered character recognition systems. Demonstrates expertise in data visualization, predictive modeling, and full-stack development to drive strategic initiatives.

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[02] Technical_Engine

[03] Technical_Skills

PYTHON · JAVA · SPRING BOOT · SQL · C · HTML · JAVASCRIPT · TENSORFLOW · KERAS · FASTAPI · SCIKIT-LEARN · CNN ARCHITECTURE · DATA PIPELINES · GIT · LM STUDIO · OPENAI · AIDER · UNITY · STREAMLIT · BOOTSTRAP 5 · MYSQL · GOOGLE EARTH ENGINE · SENTINEL-2 ANALYSIS · ESP32 TELEMETRY

[04] Soft_Skills

LEADERSHIP · STRATEGIC PROBLEM SOLVING · COMMUNICATION

[05] Experience_Log

Jan 2025 - Feb 2025

● ACTIVE

Data Science Intern

[REF: EX-001]

Ediglobe

- Engineered end-to-end analytical pipelines to solve complex business bottlenecks, contributing to quarterly strategic goal achievement.
- Optimized operational workflows by leveraging data insights, resulting in measurable enhancements to business value and process efficiency.
- Synthesized critical business intelligence for high-level stakeholders, facilitating data-driven decision-making through advanced statistical modeling.
- Developed interactive data visualizations and predictive models to interpret trends and influence strategic company-wide initiatives.

[06] Project_Schematics

PROJ_001 [LINK] github.com/Madhan1012/Veldora

Multi-Modal MRV Framework

Developed a patent-pending hybrid framework for Soil Organic Carbon (SOC) estimation using IoT and Satellite data. This system fuses Sentinel-2 multi-spectral imagery with real-time ground telemetry from ESP32 sensors, along with automated data ingestion pipelines for high-precision environmental analytics.

IOT · SATELLITE DATA · SENTINEL-2 · ESP32 · DATA INGESTION PIPELINES

PROJ_002 [LINK] github.com/Madhan1012/Character-Recognition-V2

Ancient Tamil Character Recognition

Built an AI-powered web application using TensorFlow/Keras and FastAPI to decipher 8th-century Vatteluthu inscriptions. This involved developing a CNN-based OCR model to translate ancient scripts into modern Tamil, preserving historical linguistic data, and featuring a Streamlit frontend.

TENSORFLOW · KERAS · FASTAPI · CNN · OCR · STREAMLIT

PROJ_003 [LINK] github.com/Madhan1012/Natural-Language-To-SQL

NL-to-SQL Intent Engine

Created a modular bot designed to translate natural language queries into executable SQL commands against relational datasets. This engine implemented an intent-based classification system using Scikit-learn to reduce data access friction for non-technical users.

SQL · SCIKIT-LEARN

PROJ_004 [LINK] github.com/Madhan1012/AegisLM

AegisLM

Architected a Template-LM hybrid system utilizing Java Spring Boot and Python to streamline local LLM orchestration. Engineered a complex relational database schema in MySQL with recursive joins and designed integration protocols for local inference servers, including LM Studio and Qwen 2.5 Coder.

JAVA | SPRING BOOT | PYTHON | LLM ORCHESTRATION | MYSQL | RECURSIVE JOINS | LM STUDIO | QWEN 2.5 CODER

[07] Education

Bachelor’s of Technology in ComputerScience (Data Science)
Vel Tech University
2023 - Present

12th in MPCB
Everwin Vidhyasharam
2022 - 2023